

TASK1-

For a startup team doing **software development and video conferencing at the same time**, hardware should be selected by balancing compatibility, workload, and future upgrade needs.

1. CPU Selection

- Choose a modern **multi-core CPU** (at least 6 cores/12 threads; 8 cores or more is better for heavy development).
- Check that the CPU uses a **socket supported by the motherboard** and is compatible with its chipset.
- Higher clock speeds improve tasks such as compiling code, while multiple cores help when running IDEs, browsers, virtual machines, and video-conferencing software simultaneously.
- Avoid choosing a very powerful CPU with insufficient RAM or slow storage, as these can become bottlenecks.

2. RAM Selection

- **16 GB RAM** is a practical minimum for development + video conferencing.
- **32 GB** is recommended for developers running multiple IDEs, browsers, Docker containers, virtual machines, or large projects.
- Check the motherboard's supported **DDR generation, maximum capacity, and RAM speed**.
- Use matched memory modules (for example, 2 × 16 GB) when appropriate to enable dual-channel performance.
- Insufficient RAM causes the system to use storage as virtual memory, significantly reducing performance.

3. Storage Selection

- Select an **NVMe PCIe SSD** rather than a traditional HDD for the operating system, development tools, source code, and frequently used files.
- **512 GB** is a reasonable minimum, while **1 TB or more** is better for developers working with large projects, videos, containers, and development environments.
- Confirm that the motherboard has a compatible **M.2 slot and PCIe generation**.
- Keep sufficient free space because an almost-full SSD can negatively affect performance.

- A separate larger SSD/HDD can be used for archived video files and backups.

Recommended Startup Configuration

Component	Recommended Specification	Reason
CPU	6–8+ cores, modern architecture	Handles development and conferencing simultaneously
RAM	32 GB DDR5	Prevents memory bottlenecks during multitasking
Storage	1 TB NVMe PCIe SSD	Fast booting, compiling, file access, and project storage
Motherboard	Compatible CPU socket/chipset	Ensures CPU and RAM compatibility
Upgradeability	Extra RAM/M.2 slots	Allows future expansion

TASK-2

System Resource Allocation and Hardware Bottlenecks

After deploying a resource-intensive background application such as **antivirus software**, system performance can decrease because the application continuously consumes hardware resources that are also needed by other programs.

1. CPU Usage:

Antivirus software performs real-time scanning of files, applications, and network activity. This can consume significant CPU processing power, leaving fewer CPU resources for applications such as IDEs, browsers, or video-conferencing software.

2. RAM Usage:

Antivirus services run continuously in the background and occupy RAM. If available RAM becomes low, the operating system may use virtual memory on the storage drive, which is much slower and can cause system lag.

3. Storage Bottlenecks:

Antivirus programs frequently read and scan files. On slower or heavily loaded storage devices, this creates high disk I/O and can slow down application loading, file operations, and system startup.

4. Resource Allocation:

The operating system distributes CPU time, memory, and I/O resources among

running processes. When background applications receive a large share of these resources, foreground applications may receive fewer resources and become slower.

5. **Hardware Bottlenecks:**

A system with an older CPU, limited RAM, or a slow HDD is more likely to experience noticeable performance degradation. For example, an antivirus scan running on a system with **8 GB RAM and an HDD** may cause considerably more slowdown than the same scan on a system with **32 GB RAM and an NVMe SSD**.

Example

Suppose a developer is running **Visual Studio, Chrome, a database server, and a video meeting**. If antivirus software starts a full system scan, CPU utilization and disk activity may increase significantly. The other applications then have fewer resources available, resulting in **slower compilation, delayed application responses, and video-call interruptions**.

TASK-3

Systematic Troubleshooting Process

When a peripheral device such as a **printer, scanner, webcam, or USB device** works inconsistently across different workstations, the problem may be caused by **driver incompatibility, operating-system conflicts, or incorrect configuration**. A systematic troubleshooting process can identify and resolve the issue.

1. **Identify and Define the Problem**

Determine exactly what is failing—for example, the device works on one computer but is not detected or functions incorrectly on another. Record the affected workstations, operating-system versions, error messages, and device behavior.

2. **Check Physical Connections**

Inspect USB cables, power connections, ports, and adapters. Try a different USB port or cable to eliminate a physical connection problem.

3. **Compare Workstations**

Compare the working and non-working computers for **OS version, architecture (32-bit/64-bit), driver version, device settings, and security policies**. This helps identify what is different between the systems.

4. **Check Device Manager**

On Windows, open **Device Manager** and look for warning symbols or error codes. Remove disabled or conflicting devices and check whether Windows correctly detects the peripheral.

5. **Update or Reinstall Drivers**

Install the correct driver version designed for both the **specific device model and**

operating system. If an existing driver is corrupted or incompatible, uninstall it and perform a clean driver installation.

6. Check for OS Conflicts

Install available operating-system updates and verify that the device is supported by the OS. Check for conflicts with other drivers, security software, device-management policies, or recently installed applications.

7. Test Using Standard Drivers

If the manufacturer's driver causes problems, test the device using a compatible built-in or generic Windows driver. This helps determine whether the manufacturer's driver is responsible.

8. Restart and Test Again

Restart the workstation after driver or configuration changes. Test the peripheral with several applications to confirm that the problem has been resolved.

9. Document the Solution

Record the working driver version, OS version, configuration changes, and troubleshooting steps. Apply the verified configuration to other affected workstations.

Example

If a **USB printer works on Windows 11 but prints incorrectly on another workstation**, compare the driver versions first. If the second computer has an older or incompatible driver, uninstall it and install the manufacturer's Windows 11-compatible driver. After restarting and testing a print job, the issue can be confirmed as resolved.

TASK- 4

Role of UPS and Surge Protection

A **Uninterruptible Power Supply (UPS)** and **surge protection** help protect computers and storage devices from power problems that can cause **file-system corruption, hardware damage, and data loss**.

1. UPS provides backup power:

During a power failure, a UPS supplies battery power for a short period. This prevents the computer from shutting down suddenly while files are being written.

2. Prevents file-system corruption:

A sudden power loss during a file write or system update can leave files incomplete or damage file-system structures. A UPS allows the user or system to perform a **safe shutdown**, reducing this risk.

3. **Handles voltage fluctuations:**

Some UPS systems regulate unstable voltage, protecting equipment from **brownouts and brief power interruptions** that can affect sensitive electronic components.

4. **Surge protection prevents voltage spikes:**

A surge protector limits sudden increases in voltage caused by events such as lightning or electrical switching. This helps protect the **power supply, motherboard, SSD/HDD, and other components** from electrical damage.

5. **Supports data integrity:**

By reducing unexpected shutdowns and hardware damage, UPS and surge protection help maintain the availability and integrity of stored data.

Example

If a computer is saving an important project file when the electricity suddenly goes off, the UPS keeps the computer running temporarily. The user can save the file and shut down safely instead of losing data due to an abrupt power failure.

SECTION-B

TASK-1

A workstation should be assembled in a safe and systematic order to prevent component damage and ensure proper operation.

Installation Sequence

1. **Prepare the Workspace**

- Shut down and unplug all equipment.
- Work on a clean, dry, non-carpeted surface.
- Use an anti-static wrist strap or regularly ground yourself to prevent electrostatic discharge (ESD).

2. **Install the CPU**

- Place the motherboard on its box or an anti-static surface.
- Open the CPU socket and align the CPU with the socket markings.
- Carefully place the CPU into the socket and lock the retention mechanism.

3. **Install CPU Cooling**

- Apply thermal paste if the cooler does not have pre-applied paste.
- Position the CPU cooler over the CPU and secure it according to the manufacturer's instructions.

- Connect the cooler's fan cable to the motherboard's **CPU_FAN** header.

4. Install RAM

- Open the appropriate RAM slots.
- Align the notch on each RAM module with the slot.
- Press down firmly until the retaining clips lock the modules into place.

5. Install Storage

- For an NVMe SSD, insert it into the appropriate **M.2 slot** and secure it with the mounting screw.
- For a SATA SSD/HDD, mount the drive in its drive bay and connect the SATA data and power cables.

6. Install the Motherboard

- Install the correct motherboard standoffs in the case.
- Carefully place the motherboard inside the case and secure it with the appropriate screws.

7. Install the Power Supply (PSU)

- Place the PSU in its designated position.
- Secure it with screws.
- Connect the required motherboard, CPU, storage, and graphics-card power cables.

8. Install the Graphics Card

- Remove the appropriate expansion-slot covers.
- Insert the graphics card into the correct PCIe slot.
- Secure it to the case and connect its additional power cable if required.

9. Connect Front-Panel and Case Cables

- Connect the power switch, reset switch, USB, audio, and indicator LED cables to the motherboard.
- Connect case fans to the appropriate fan headers.

10. Final Inspection

- Check that all components are firmly installed.

- Ensure cables are not obstructing fans.
- Verify that no loose screws or unused cables are causing problems.
- Confirm that the CPU cooler and RAM are properly seated.

11. First Boot and Testing

- Connect the monitor, keyboard, and power cable.
- Turn on the workstation and enter the BIOS/UEFI.
- Verify that the **CPU, RAM, and storage drive** are detected.
- Check CPU temperature and fan operation.
- Install the operating system and required drivers after successful hardware detection.

Component Installation Order

Workspace preparation → CPU → CPU cooler → RAM → Storage → Motherboard → PSU → Graphics card → Cables → Final inspection → BIOS/UEFI testing → Operating system

Result: Following this sequence reduces the risk of ESD damage, incorrect connections, and component installation errors while providing a clear record for troubleshooting and future upgrades.

TASK-2

Troubleshooting and Resolving a RAM Mismatch Error in BIOS/UEFI

A **RAM mismatch error** occurs when the BIOS/UEFI detects a problem with the installed memory, such as incompatible modules, incorrect settings, or improperly seated RAM.

Troubleshooting Steps

1. Shut Down the Computer

- Turn off the workstation and disconnect the power.
- Press the power button for a few seconds to discharge remaining power.

2. Check RAM Installation

- Open the case and ensure the RAM modules are properly seated.
- Remove and reinstall the modules until the locking clips click into place.
- Check the motherboard manual for the recommended slots, especially when using two modules.

3. Check RAM Detection in BIOS/UEFI

- Start the computer and enter BIOS/UEFI using the appropriate key, such as **Delete, F2, or F10**.
- Check the reported total memory.
- Compare the detected capacity with the physically installed RAM.

4. **Verify RAM Specifications**

- Check whether the modules have compatible **DDR generation, capacity, speed, and voltage**.
- For example, DDR4 RAM cannot be installed in a DDR5-only motherboard.

5. **Reset Memory Settings**

- If an incorrect memory profile is enabled, disable **XMP/EXPO** temporarily.
- Load **BIOS/UEFI optimized/default settings** and restart.
- This allows the system to test the RAM using safer default settings.

6. **Test RAM Modules Individually**

- If the problem remains, shut down the computer.
- Test one RAM module at a time in the recommended slot.
- Repeat the test with the other module to identify a defective module or slot.

7. **Update BIOS/UEFI if Required**

- Check the motherboard manufacturer's supported memory list and BIOS updates.
- If the installed RAM is supported but not correctly detected, updating the BIOS/UEFI may improve memory compatibility.
- Follow the manufacturer's update procedure carefully.

8. **Run a Memory Test**

- After the BIOS correctly detects the RAM, run **Windows Memory Diagnostic** or another reliable memory-testing utility.
- Check for errors before considering the problem resolved.

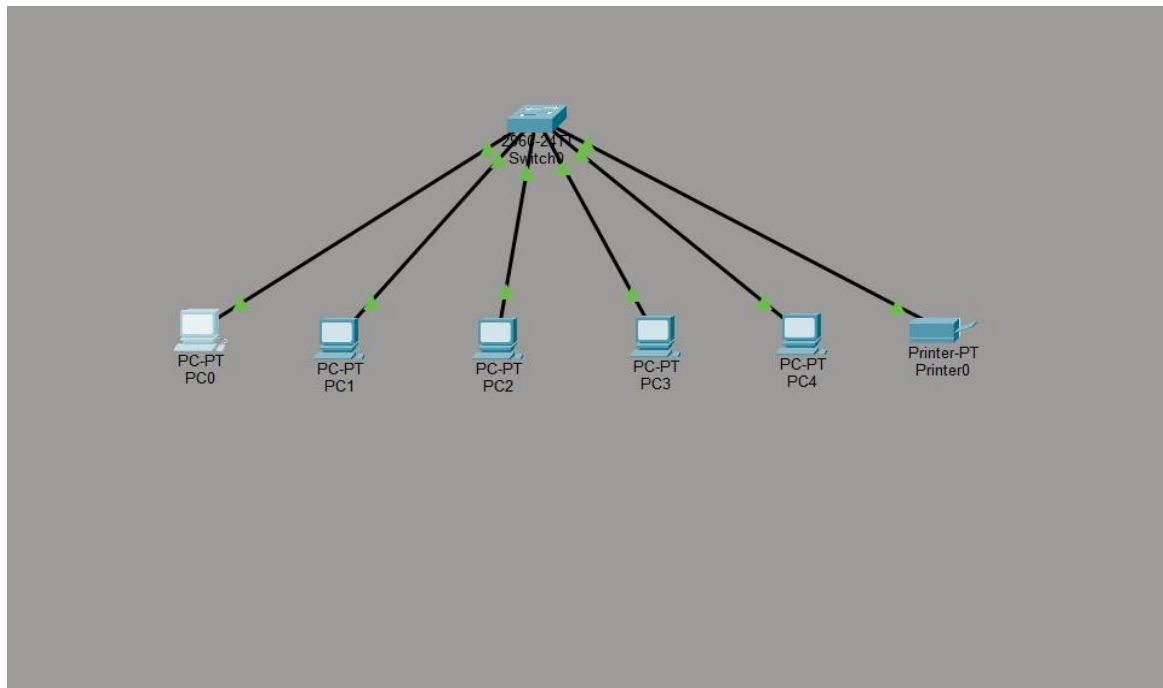
TASK-3

Q-Build a LAN topology in packet tracer for 5 systems and 1 printer using a class c subnet.

To design and configure a Local Area Network (LAN) in Cisco Packet Tracer by connecting five computers and one printer using a Class C IP addressing scheme, and to verify network connectivity.

Network Topology

The network consists of one switch connected to five computers and one printer. All devices are connected using Copper Straight-Through cables.



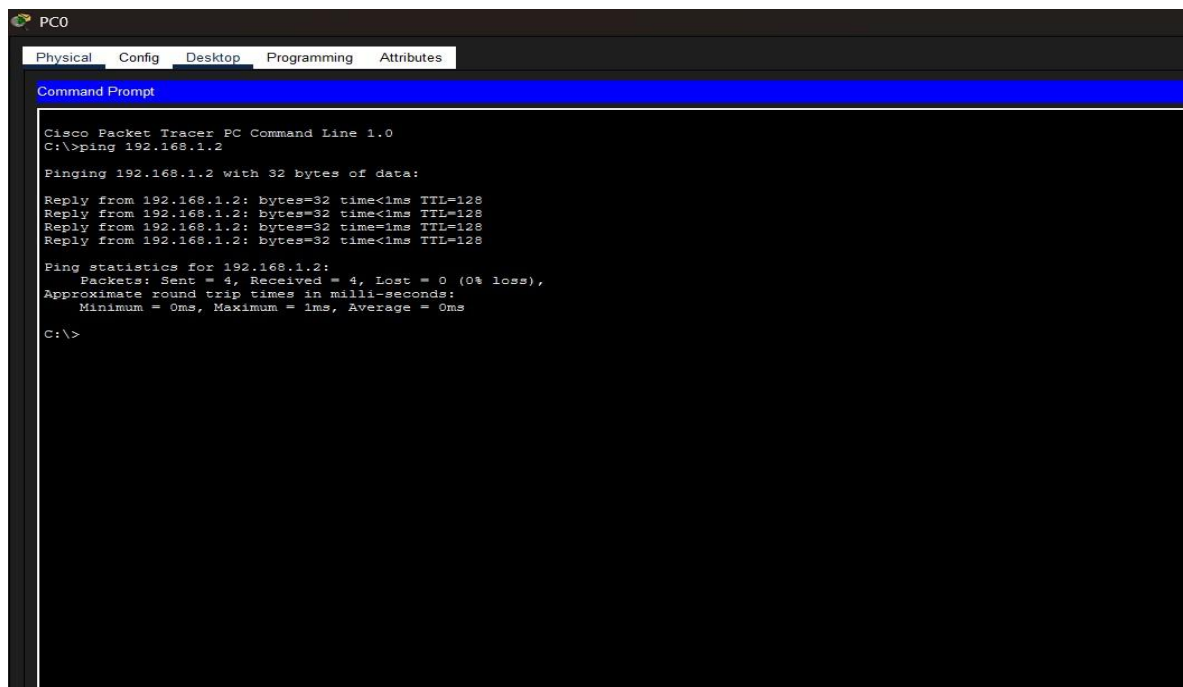
Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.10	255.255.255.0

Network Address: 192.168.1.0/24

Default Gateway: Not Required (No Router Used)

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place one Cisco 2960 Switch in the workspace.
3. Add five PCs and one Printer from the End Devices section.
4. Connect all devices to the switch using Copper Straight-Through cables.
5. Configure the IP address and subnet mask for each device according to the IP addressing table.
6. Save the configuration.
7. Open the Command Prompt on PC0 and test connectivity using the **ping** command with the IP addresses of the other PCs and the printer.
8. Verify that all devices respond successfully, confirming proper LAN communication.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Observation

- All devices were assigned unique IP addresses within the Class C network (192.168.1.0/24).
- The switch successfully connected all devices.
- Ping tests between the computers and the printer were successful.
- The LAN operated correctly without requiring a router because all devices were on the same subnet.

The LAN topology consisting of five computers and one printer was successfully created in Cisco Packet Tracer using a Class C subnet (192.168.1.0/24). All devices communicated successfully through the switch, and the network connectivity was verified using the **ping** command.

TASK-4

1. Prepare for Installation

1. Back up important files to an external drive or cloud storage.
2. Create a bootable **Windows/Linux installation USB**.
3. Verify that the workstation meets the OS hardware requirements.
4. Enter **BIOS/UEFI** and confirm that the installation USB is detected.
5. Set the USB as the temporary boot device.
6. If required, configure **UEFI boot mode** rather than Legacy/CSM.

2. Start the Clean Installation

1. Boot from the installation USB.
2. Select language, region, keyboard, and time settings.
3. Choose **Install**.
4. Select the appropriate edition when prompted.
5. Choose **Custom/Advanced installation** to perform manual partitioning.

3. Manual Disk Partitioning

For a **500 GB SSD**, an example partition layout is:

Partition	Example Size	Purpose
EFI System Partition	100–300 MB	UEFI boot files
Recovery	~500 MB–1 GB	Recovery environment
OS (C:)	150–200 GB	Operating system and applications
Data (D:)	Remaining space	Documents/projects

Partition sizes are examples. The exact layout depends on the OS and whether the system uses UEFI/GPT.

During a true clean installation, verify the correct disk **before deleting or formatting partitions**, because these operations can permanently erase data.

4. Install the Operating System

1. Select the intended OS partition.
2. Format it if necessary for a clean installation.
3. Start the installation.
4. Allow the computer to restart as required.
5. Remove the USB when the system begins booting from the internal drive.

5. Post-Installation Configuration

After the first successful boot:

- Install motherboard/chipset, graphics, network, and other required drivers.
- Run operating-system updates.
- Enable the built-in firewall and security features.
- Set a strong account password/PIN.
- Configure automatic updates.
- Verify that Device Manager has no unknown devices or driver errors.

6. Startup Optimization

Open **Task Manager** → **Startup apps** in Windows and review applications that launch automatically.

Disable unnecessary startup programs such as:

- Unneeded update assistants
- Applications that are rarely used
- Unnecessary background launchers

Do **not** disable essential security software, hardware drivers, or system components without understanding their purpose.

Also:

- Uninstall unnecessary applications.
- Keep adequate free space on the system drive.
- Avoid installing excessive background utilities.
- Check **Task Manager** → **Startup apps** periodically after installing new software.

7. Test and Document

Record:

- OS version and installation date
- Disk/partition layout
- Driver versions
- Startup applications disabled
- Boot time before and after optimization
- Any installation or driver issues encountered

Result: A clean OS installation with a planned partition structure provides a stable system, while startup optimization reduces unnecessary background activity and can improve boot time and overall responsiveness.

SECTION- C

Sure — here is a **slightly changed version** of the case study answers while keeping the same technical meaning and making it suitable for submission.

Case Study Report: Troubleshooting Random Workstation Restarts Under Heavy Load

Scenario

At **Nexus Innovations**, three rendering workstations pass POST normally in the morning but randomly restart during heavy rendering tasks later in the day. The systems are installed inside enclosed desk cabinets with limited ventilation. This causes heat to accumulate and affects system stability.

Objective

To identify the likely hardware causes of unexpected restarts, understand why the problem occurs under heavy workloads, use stress-testing methods to confirm the fault, recommend solutions, and determine whether a PSU upgrade is required.

1. Hardware Components Most Likely Responsible

A. CPU

The CPU performs intensive calculations during rendering and produces considerable heat. If its temperature becomes too high, the system may automatically shut down or restart as a safety mechanism.

Possible symptoms:

- CPU temperature becomes unusually high.
- Restarts occur mainly during rendering.
- CPU performance decreases because of thermal throttling.

B. Power Supply Unit (PSU)

Rendering increases the power demand of the CPU, GPU, RAM, and other components. A weak, aging, or faulty PSU may fail to deliver stable power when the workload increases.

Possible symptoms:

- Random restarts during demanding tasks.
- System becomes unstable under high power consumption.
- Sudden shutdowns without a warning message.

C. Cooling and Case Fans

The CPU cooler and case fans are responsible for removing heat from the system. Dust, defective fans, old thermal paste, or restricted cabinet airflow can cause temperatures to rise rapidly.

Possible symptoms:

- Excessive internal temperature.
- Fans running loudly or continuously.
- System crashes or restarts after extended use.

2. Why the Systems Pass POST in the Morning but Fail Later

POST mainly checks whether essential hardware is functioning correctly when the computer starts. In the morning, the workstations are relatively cool and have not yet experienced prolonged heavy workloads.

Later in the afternoon:

- Rendering places continuous load on the CPU and GPU.
- Heat builds up inside the enclosed cabinets.
- Limited ventilation prevents hot air from escaping efficiently.
- Internal component temperatures gradually increase.
- Thermal protection may eventually cause the system to restart.

Therefore, the computers can successfully complete POST while still developing a **thermal stability problem during sustained workloads**.

3. Using Stress Testing to Identify the Problem

A stress test can reproduce heavy workloads and help determine whether overheating or power delivery is responsible.

Procedure

Step 1: Install a suitable stress-testing tool such as **OCCT, Prime95, AIDA64, or Cinebench**.

Step 2: Install a monitoring tool such as **HWiNFO or HWMonitor**.

Step 3: Record the initial:

- CPU temperature
- GPU temperature
- Fan speed
- CPU clock speed
- Relevant voltage readings

Step 4: Run the CPU/GPU stress test for approximately **20–30 minutes**, while continuously monitoring temperatures and system behavior.

Step 5: Analyze the results.

If temperatures rise excessively, the system throttles, or the workstation restarts during the test, this provides strong evidence of a **thermal or power-delivery problem**. A single temperature reading should not be used as the only diagnosis; component specifications and the actual failure behavior should also be considered.

4. Recommended Solutions

Solution 1: Improve Cabinet Ventilation

Move the workstations out of enclosed cabinets or provide sufficient space for fresh air to enter and hot air to escape.

Benefits:

- Better circulation of cool air.
- Reduced internal temperatures.
- More effective operation of CPU/GPU cooling.

Solution 2: Maintain the Cooling System

Perform regular hardware maintenance by:

- Removing dust from heatsinks and fans.
- Checking that all fans are working.
- Replacing thermal paste when necessary.
- Improving cable management to avoid blocking airflow.
- Adding suitable case fans if required.

Solution 3: Check Other Hardware

If the problem continues after improving airflow, test the:

- PSU
- RAM

- CPU cooler
- Motherboard
- GPU

Replace components only when testing indicates that they are faulty or unsuitable.

5. Is a PSU Upgrade Required?

A PSU upgrade should **not automatically be recommended** just because the computer restarts during rendering.

PSU Upgrade Is Recommended If:

- The PSU's rated capacity is below the system's actual power requirements.
- The PSU is old, damaged, or unreliable.
- Voltage/power delivery becomes unstable under load.
- A high-power GPU or CPU was recently added.
- The PSU does not provide adequate connectors or sufficient capacity for the configuration.

A quality PSU with sufficient capacity can provide more reliable power during demanding workloads.

PSU Upgrade Is Not Required If:

- The existing PSU has adequate capacity.
- Power delivery remains stable during testing.
- The restart disappears after improving ventilation.
- Monitoring confirms that overheating was the primary cause.

In this case, spending money on a new PSU would not address the main problem.

Final Observation

The evidence indicates that **restricted airflow is the primary suspected cause** of the random restarts. The workstations pass POST when they are cool, but prolonged rendering causes heat to accumulate inside the poorly ventilated cabinets.

Stress testing and hardware monitoring can be used to reproduce the failure and confirm the temperature and power conditions. The first corrective action should therefore be to **improve ventilation and service the cooling system**. The PSU should be upgraded only if testing shows that its capacity or power delivery is inadequate.

Final Result: After improving airflow, cleaning the cooling components, checking hardware, and verifying PSU performance, the workstations should be able to complete heavy rendering tasks more reliably without unexpected restarts.